

Motivation

- Basal water controls ice sheet sliding and therefore ice discharge to the ocean.
- Ice-shelf hydrology can weaken buttressing and affect ice sheet stability.
- Periodic subglacial lake drainage can cause transient accelerations of ice flow.
- Freshwater discharge influences ocean circulation and ice-shelf melting.
- Hydrology is difficult to observe directly, making numerical models an essential tool.

Ice flow modelling

- Ice thickness evolves following mass conservation:
$$\frac{\partial H}{\partial t} = -\nabla \cdot (H\bar{\mathbf{u}}) + \dot{a}.$$
- A stress balance equation determines the ice velocity $\bar{\mathbf{u}}$.
- The basal shear stress $\tau_b(\bar{\mathbf{u}}, N)$ appears in this equation, which depends on the effective pressure $N = P_{\text{ice}} - P_{\text{water}}$.
- Subglacial hydrology determines P_{water} , thereby controlling basal resistance.
- An example of a sliding law is the regularized Coulomb law:

$$\tau_b = C \frac{|\mathbf{u}_b|^{m-1} \mathbf{u}_b}{(|\mathbf{u}_b| + u_0)^m} N.$$

- Lower effective pressure (higher water pressure) reduces basal friction and increases ice flow speed.

Ice sheet hydrology



Credit: Laura A. Stevens

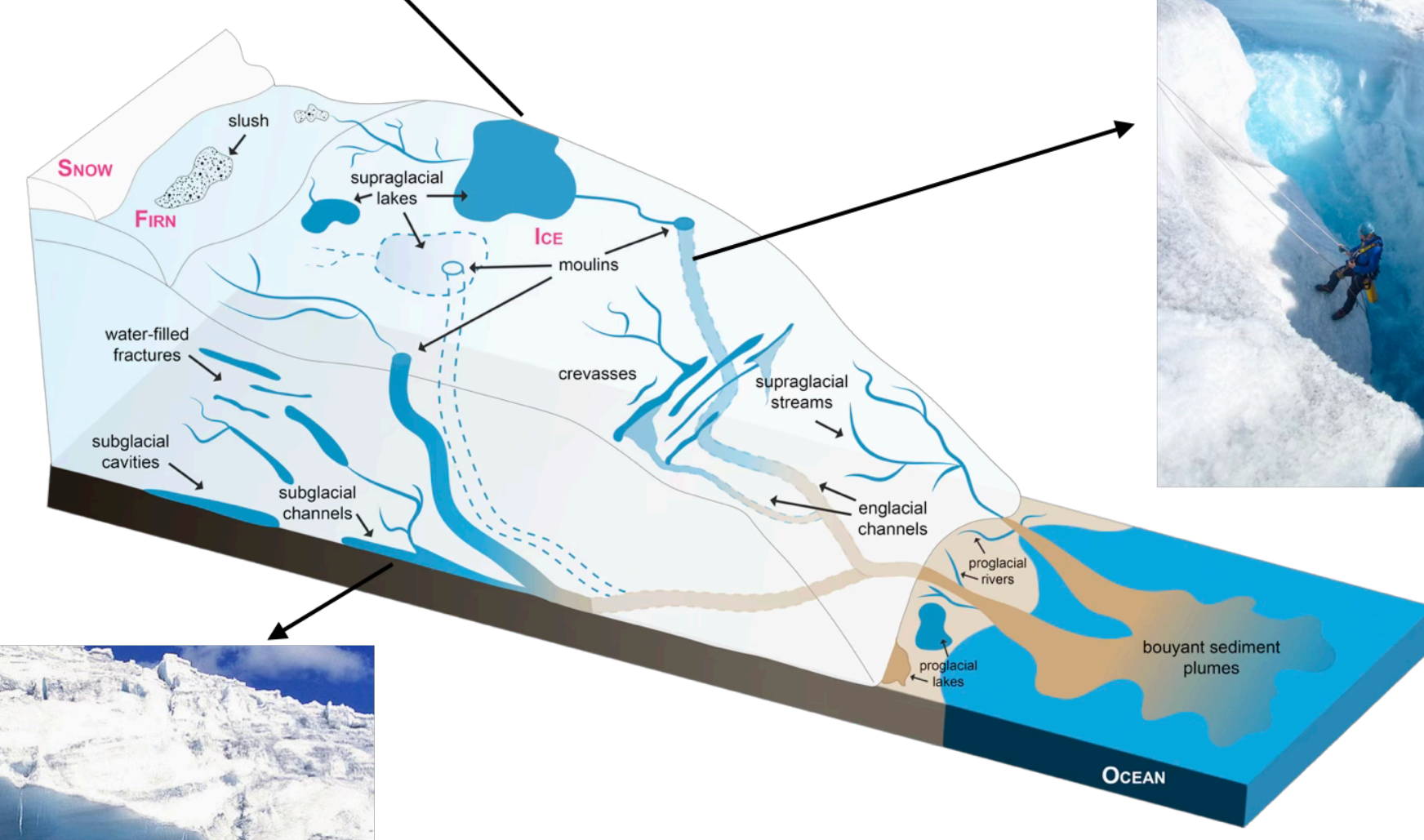
Credit: Henry Patton



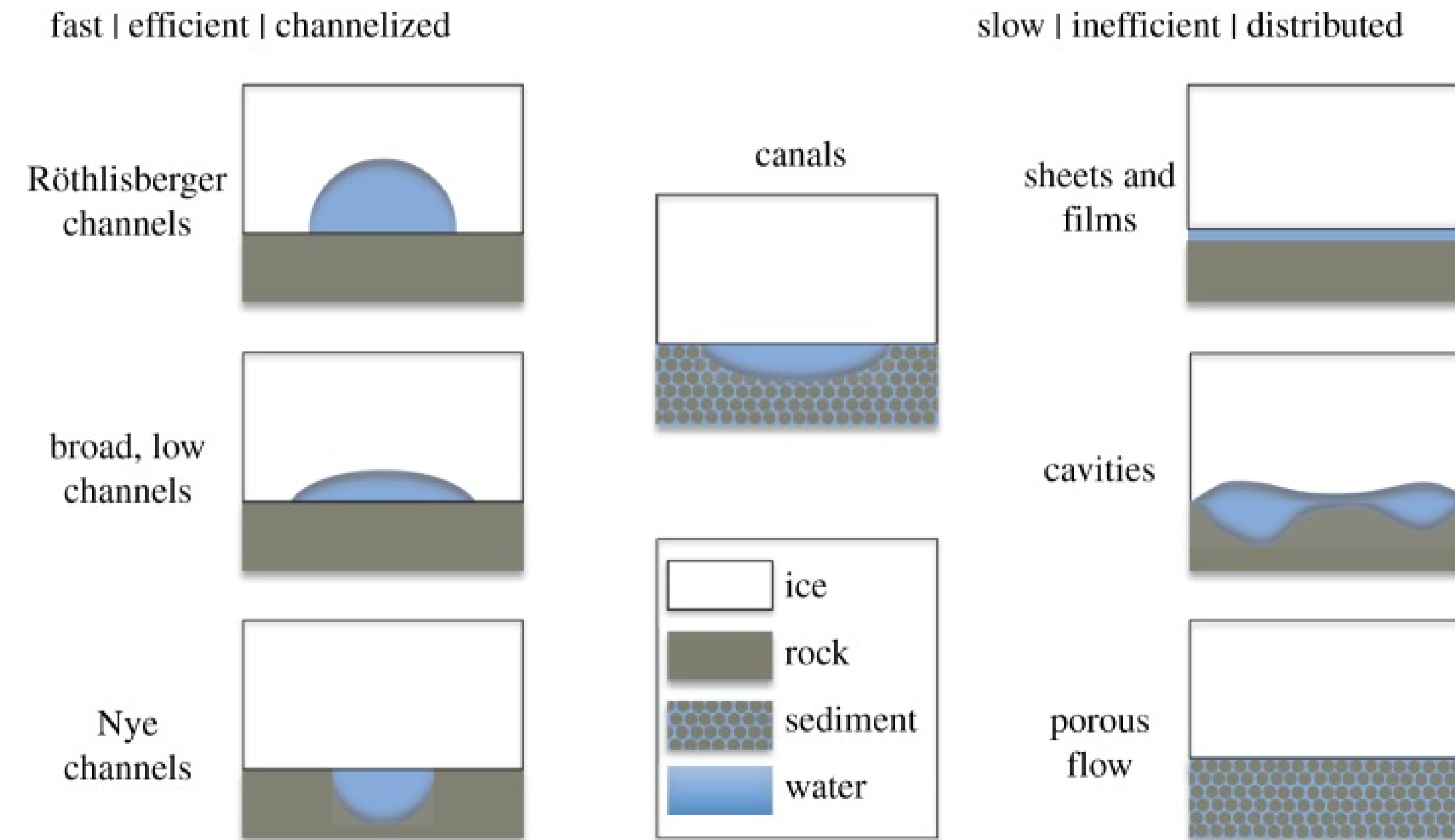
Credit: M. J. Hambrey



Credit: Agnieszka Gautier



Ice sheet hydrology modelling



SHAKTI transient hydrology model

- Water conservation:
$$\frac{\partial b}{\partial t} = -\nabla \cdot \mathbf{q} + \frac{\dot{m}}{\rho_w} + i_{e \rightarrow b}.$$
- Cavity size also changes through melt opening, sliding over bed roughness, and viscous creep closure:
$$\frac{\partial b}{\partial t} = \frac{\dot{m}}{\rho_i} + \beta |\mathbf{u}_b| - A |N|^{n-1} N b.$$
- Basal melt rate $\dot{m}$ is generated by geothermal heating G , frictional heating from sliding, and dissipation of flowing water.

- Water flux is driven by gradients in hydraulic head $h = P_{\text{water}}/(\rho_w g) + z_{\text{bed}}$:

$$\mathbf{q} = \frac{b^3 g}{12\nu(1 + \omega Re)} (-\nabla h).$$

- We can solve for h , which gives P_{water} , and hence obtain N using $P_{\text{ice}} = \rho_i g H$.

Kori-ULB steady-state hydrology model

- Water conservation with steady state cavity size:

$$\nabla \cdot \mathbf{q} = \frac{\dot{m}}{\rho_w}.$$

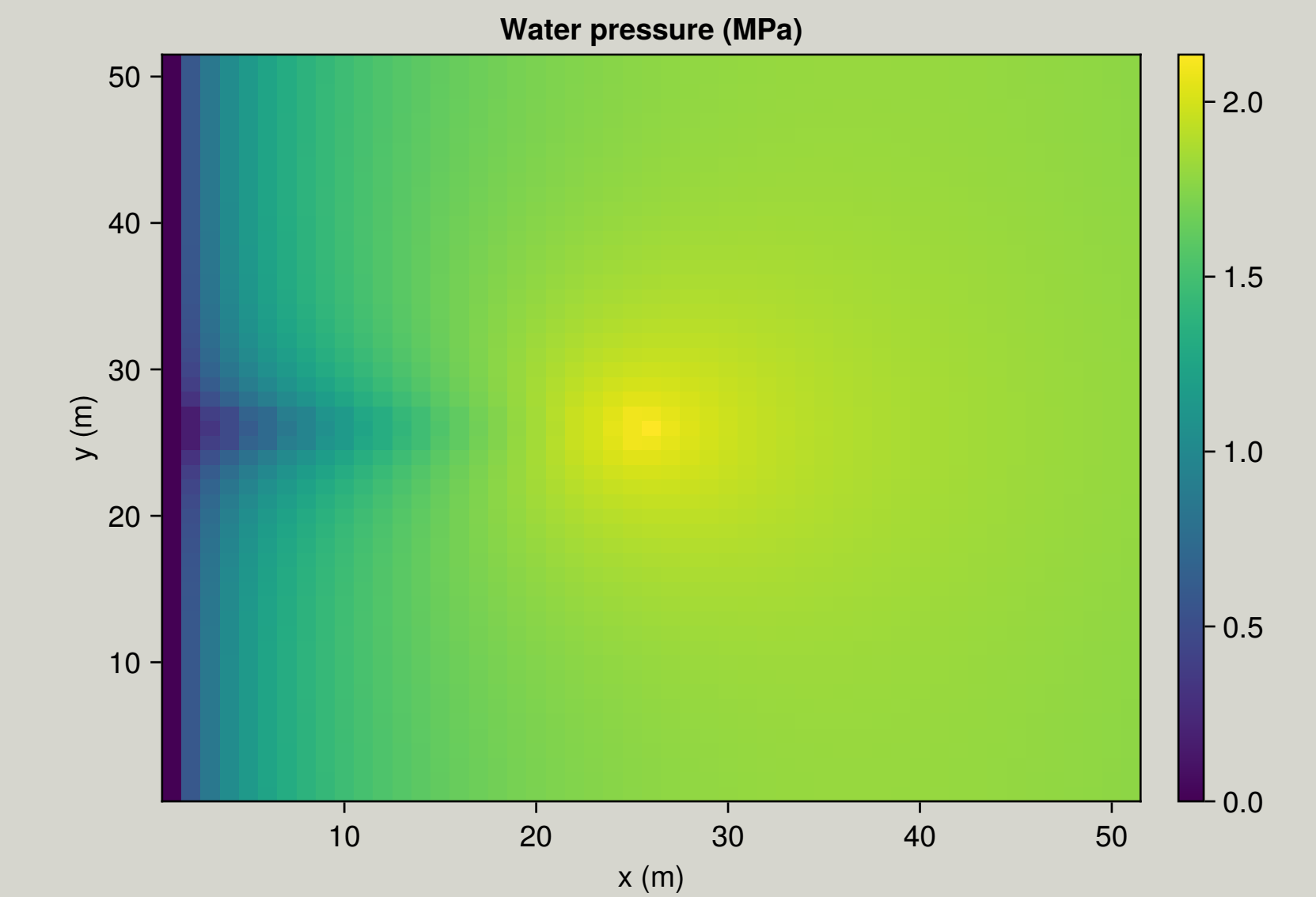
- Approximate the hydraulic potential by the geometric potential $\rho_w g \nabla h \approx \nabla \phi_0$.
- The conduit discharge is represented using a Darcy–Weisbach relation:

$$\mathbf{q} = K S^{5/4} |\nabla \phi_0|^{-1/2} (-\nabla \phi_0).$$

- Analytical expressions can then be obtained for conduit area S and effective pressure N .

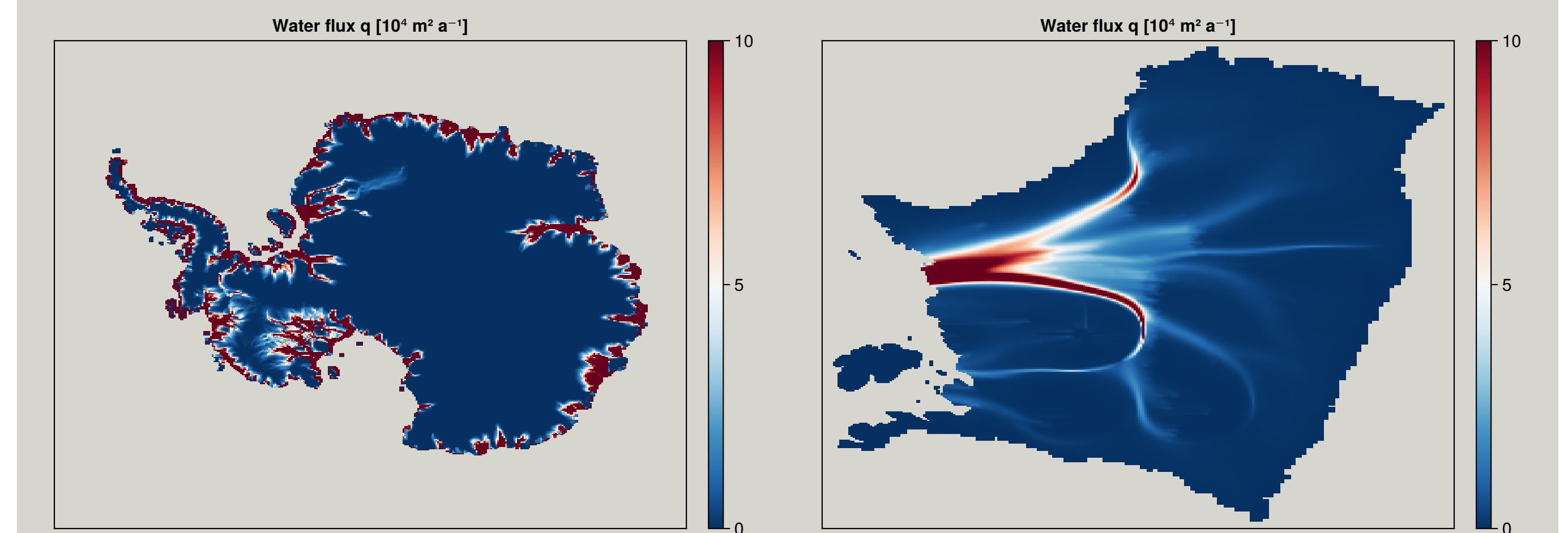
SHAKTI hydrology model results

- Water injected at the moulin initially spreads through the distributed drainage system.
- As discharge increases downstream, an efficient channel forms.
- The channel transmits large water fluxes at lower pressure and draws water from the surrounding distributed system.



Kori-ULB hydrology model results

- Water flux follows major ice streams and converges into efficient drainage pathways.
- The model reproduces large-scale drainage patterns across Antarctica.
- Beneath Thwaites Glacier, distributed flow transitions into channelized drainage near the grounding line.



Outlook

- Perform fully coupled ice sheet and hydrology simulations for Antarctica and Greenland.
- Assess the impact of hydrology on ice dynamics and sea-level projections.
- Investigate model sensitivity to spatial resolution, sliding law, and other parameters.
- Develop strategies for calibration and validation using observations.